

Technical Datasheet — Power Tool Battery Cell

Case ID: t3_r1_mimo
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Parameter	Value	Notes
Cell Chemistry	NMC811 / Graphite Li-ion	Chen2020 parameterization
Rated Capacity	2.30 Ah	1C discharge, 25°C
Nominal Voltage	3.84 V	Midpoint voltage from simulation
Voltage Window	2.5 – 4.2 V	Chen2020 limits
Rated Energy	8.34 Wh	Integration of $V \cdot I \cdot t$
Energy Density (gravimetric)	308.4 Wh/kg	Contract-caliber (excl. electrolyte)
Energy Density (volumetric)	712.3 Wh/L	Contract-caliber (excl. electrolyte)
Maximum Continuous Discharge	5C (11.5 A)	96.8% capacity retention vs 1C
Fast-Charge Capability	4C at 45°C	No lithium plating (anode min +0.005V)
Maximum Temperature (4C charge)	54.5°C (327.6 K)	Thermal limit 60°C: PASS
Power Density	65,280 W/kg	$V_{OC}^2 / (4 \cdot DCR) / \text{mass}$
DC Resistance	2.35 mΩ	At 10% discharge
Operating Temperature	25°C (simulated)	Simulation conditions
Cycle Life	Not simulated	Requires aging model — not evaluated
Cell Mass	27.04 g	Excl. electrolyte
Cell Dimensions	65 × 1580 × 0.114 mm	Height × Width × Total Thickness
Safety — Plating	No plating at 4C	Anode potential ≥ 0V throughout
Safety — Thermal	T_max 54.5°C at 4C charge	Below 60°C limit

Application Suitability

This cell is designed for **power tool applications** requiring:

- High rate discharge (5C continuous with 96.8% retention)
- Fast charging (4C without lithium plating)
- Thermal safety under aggressive operating conditions

The enhanced electrolyte transport ($t_{H^+}=0.45$, $\sigma=1.5$ S/m) and aggressive thermal management ($h=50$ W/m²K) are critical design features enabling 4C fast charge capability.